

SNN AER Output Interface

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a) Mean aggregate spike rate

$$R = Nf$$

$$R = 1024 \times 50 = 51,200 \text{ spikes/s}$$

b) Mean AER bandwidth

Packet size:

$$10 + 6 + 4 = 20 \text{ bits/packet}$$

$$B = R \times 20$$

$$B = 51,200 \times 20 = 1,024,000 \text{ bits/s}$$

$$B = \frac{1,024,000}{10^6} = 1.024 \text{ Mbit/s}$$

c) Interface comparison

SPI:

$$50 \text{ Mbit/s} > 1.024 \text{ Mbit/s}$$

Can sustain the mean rate.

I²C:

$$3.4 \text{ Mbit/s} > 1.024 \text{ Mbit/s}$$

Can sustain the mean rate.

AXI4-Lite:

$$100 \text{ Mbit/s} > 1.024 \text{ Mbit/s}$$

Can sustain the mean rate.

All three interfaces can sustain the mean rate. The lowest-complexity interface that suffices is I²C.

d) Burst bandwidth and buffering

Burst spikes:

$$N_{\text{burst}} = 0.25 \times 1024 = 256 \text{ spikes}$$

Burst packet data:

$$256 \times 20 = 5120 \text{ bits}$$

Peak bandwidth over 1 ms:

$$B_{\text{burst}} = \frac{5120}{1 \times 10^{-3}} = 5.12 \times 10^6 \text{ bits/s}$$
$$B_{\text{burst}} = 5.12 \text{ Mbit/s}$$

Burst-to-mean ratio:

$$\frac{B_{\text{burst}}}{B_{\text{mean}}} = \frac{5.12}{1.024} = 5$$

I²C cannot sustain the instantaneous burst:

$$3.4 \text{ Mbit/s} < 5.12 \text{ Mbit/s}$$

Bits transmitted by I²C during the 1 ms burst:

$$3.4 \times 10^6 \times 10^{-3} = 3400 \text{ bits}$$

Backlog:

$$5120 - 3400 = 1720 \text{ bits}$$
$$\frac{1720}{20} = 86 \text{ packets}$$

Minimum smoothing FIFO: about 86 packets. Safer full-burst FIFO: at least 256 packets.

e) Frame-based comparison

Frame sample rate:

$$\frac{1}{1 \text{ ms}} = 1000 \text{ samples/s}$$

Frame bandwidth:

$$B_{\text{frame}} = 1024 \times 1000 = 1,024,000 \text{ bits/s}$$
$$B_{\text{frame}} = 1.024 \text{ Mbit/s}$$

AER-to-frame ratio:

$$\frac{B_{\text{AER}}}{B_{\text{frame}}} = \frac{1.024}{1.024} = 1$$

Crossover:

$$20Nf = 1000N$$

$$20f = 1000$$

$$f_{\text{crossover}} = 50 \text{ Hz}$$

AER is the right choice when average firing rate is below the crossover rate; frame-based readout becomes competitive when activity is dense or highly synchronous.